

20 The profit made by a shop increases each year.

The profit made by the shop in year n is $\text{£}P_n$

Given that the profit made by the shop in the next year is $\text{£}P_{n+1}$ then

$$P_{n+1} = aP_n + 800 \text{ where } a \text{ is a constant.}$$

The table shows the profit made by the shop in 2018 and in 2019

Year	2018	2019
Profit	£24 000	£29 600

Work out the profit predicted to be made by the shop in 2021

£

(Total for Question 20 is 4 marks)